Rajiv Ranganath

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Career Overview

Final-year Computer Science student (CGPA 9.36, graduating May 2027) who builds AI systems and proves they work: LLM pipelines that check every answer against its source, tested Python backends, and real-time tracking software. AI Intern at IIT Kharagpur. Available to start immediately, full-time or as an intern.

Education

B.N.M. Institute of Technology 2023 – 05/2027
B.E. Computer Science & Engineering Bangalore, Karnataka
CGPA: 9.36

Sri Kumarans PU College 2021 – 2023
Class XII (PCM), State Board · 92.3% Bangalore, Karnataka

Experience

AI Intern · IIT Kharagpur · Jul 2026 to Present · Remote
- Built an LLM pipeline that extracts figures from company sustainability reports and certifies a value only when it traces back to its source; certified values match official XBRL filings **97.5% of the time (dev) and 93.3 to 100% on three of four locked holdout sets**.
- Audited the inherited evaluation code, removed **fabricated data and answer-key leakage**, and rebuilt scoring so every answer must cite its source.
- Added automatic failover across five LLM providers (Groq, Gemini, NVIDIA NIM, OmniRoute, Ollama) with run-time model discovery, covered by **337 pytest tests** on GitHub Actions CI.

Deep Learning Intern · E2E Management Services · Jun 2025 · Bangalore
- Built a diabetic retinopathy screening model by fine-tuning **ResNet-152** with a custom classifier head on the APTOS 2019 retina dataset, grading fundus images into five severity levels (No DR to Proliferative).
- Built a Streamlit app that summarises legal documents with an LLM (Groq), **flags GDPR risks** and key clauses, answers questions over the document, and tracks GDPR recitals with email and Google Sheets alerts.

Projects

LedgerGuard: Self-Verifying Payment Reconciliation (Razorpay AI Buildathon)
- Built a reconciliation engine that matches orders, payments, refunds, settlements and bank records in exact integer paise against six consistency rules, resolving **54 of 85 held-out cases with no AI**.
- An LLM investigates only the cases code cannot prove, and an evidence gate re-checks every claim against the records: accuracy rose from **89.4% to 97.7%**, and wrong closures on 30 adversarial cases fell from **25 to 0**.
- Stack: Python · FastAPI · Pydantic · httpx · pytest (59 tests)

Orbit Guard AI: Real-Time Space Debris Tracking & Collision Warning System
Tracks space debris from sparse ground-station observations and warns of likely collisions.
- **200-particle Ensemble Kalman Filter** that estimates orbits from noisy observations, modelling Earth's oblateness (J2), atmospheric drag and solar radiation pressure, with the integrator verified against two-body physics (~10^{-12} energy and momentum drift over 24 h).
- Built two observation-to-object matchers (Hungarian nearest-neighbour for clean data, JPDA for cluttered radar data) and collision alerts, streamed live to a 3D web view.
- Rebuilt the matcher stress test as a seeded 100-scenario benchmark with 95% CIs: in 8-object clutter JPDA's association rate was **6.7 points higher than GNN's (higher in 94 of 95 runs)**, and a ghost-match audit showed half of its matches were clutter.
- Stack: Python · NumPy · FastAPI · WebSocket · React · Three.js · Live demo

Cooperative MARL for Energy-Harvesting Sensor Scheduling
- Trained three cooperative multi-agent RL methods (IQL, VDN, QMIX) to schedule four solar-powered sensor nodes that each act on their own policy with no central controller, tested against seven hand-written baselines on fixed 30-seed test splits.
- Compressed policies to int8 for microcontrollers and caught a hidden **21% reward loss despite 99.7% action agreement**; fixed it by calibrating on recorded trajectories.
- Showed that a simple threshold rule still beats the best learned policy (by 4.95 reward, 95% CI [1.46, 8.44]) and reported it rather than hiding it.
- Stack: Python · PyTorch · EPyMARL · ONNX · pytest (76 tests)

Skills

- Languages: Python, SQL, C, Java
- Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, OpenCV, Pandas, NumPy, Matplotlib; CNNs and transfer learning (ResNet); state estimation (Kalman / ensemble filters), multi-target tracking
- Core CS (coursework): Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- Generative AI & LLM Systems: RAG, LLM evaluation & benchmarking, FAISS, LangChain, Groq API, Hugging Face Transformers, Sentence Transformers
- Backend & Infrastructure: REST APIs (FastAPI), WebSocket, PostgreSQL, MongoDB, Docker, Git, AWS basics (EC2, S3, Lambda, IAM), React / Three.js
- RL & Testing: PPO, QMIX, ONNX, pytest, GitHub Actions

Certifications

- Introduction to Internet of Things, NPTEL (IIT Kharagpur) · Elite + Gold, top 5% · 93%